
Effects of Sleep Deprivation on Serotonin Function in Depression

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There is considerable evidence that antidepressant treatments enhance serotonin (5-HT) function. In order to evaluate whether sleep deprivation (SD) produces alterations in 5-HT function, the increase in prolactin (PRL) produced by intravenous tryptophan (TRP) was assessed in depressed patients following SD and undisturbed sleep (US). Eleven depressed patients received mood ratings and TRP infusions after either SD or US, 1 week apart. In five women, but not six men, the TRP-induced PRL rise was markedly enhanced after SD compared to US. Mood score changes were not significantly different between US and SD and there was no significant relationship of mood changes to the TRP induced PRL response. The data suggests that SD produces an increase in 5-HT function in female depressed patients. The lack of identified SD-induced changes in 5-HT function in men may be due to lower sensitivity of the TRP-induced PRL rise in depressed men than women.

Key Words: Antidepressant treatment, depression, prolactin, tryptophan, serotonin, sleep deprivation

Introduction

Sleep deprivation for an entire night (total SD) is followed, over the next day, by an improvement in mood in two-thirds of patients with endogenous depression and about one-half of all depressed patients (Wu and Bunney 1990; Pflug and Tölle 1971; Bhanji and Roy 1975; van den Burg and van den Hoofdakker 1975; Kuhs and Tölle 1991). A marked improvement in mood typically begins in the early morning hours, and continues through the morning and early afternoon. The remission generally ends when the patient awak-

ens from the sleep period following SD (Wu and Bunney 1990), however.

Studies of neurobiological mechanisms underlying the therapeutic effect of SD on mood in depressed patients have included examination of specific sleep stage deprivations (Vogel et al 1968), measurement of the concentration of urinary and cerebrospinal fluid (CSF) monoamine neurotransmitters and their metabolites (Gerner et al 1979; Post et al 1976), and evaluation of interactions between SD and antidepressant medications (Elsenga and van den Hoofdakker 1982; Wirz-Justice et al 1976). Restriction of one or more of the electroencephalographic sleep stage(s) may be essential for mood improvement after SD (Vogel et al 1968) and may signal the importance of monoamines whose function is associated with that sleep stage. SD can be confined (Dement and Greenberg 1966) to the first or second segment of the sleep period (early partial SD or late partial SD,

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respectively), or to a specific electroencephalographic (EEG) sleep phase (Monti 1983; Agnew et al 1964), such as rapid eye movement (REM SD). Total SD, late partial SD (Sack et al 1988), and REM SD appear to be more effective than other types of partial SD at improving mood in melancholia or endogenous depression (Wu and Bunney 1990; Vogel et al 1968; Schilgen and Tölle 1980; Gillin et al 1989; Reynolds et al 1987; Giedke et al 1990). Even a hypothetically "pure" REM disruption also diminishes deep non-REM sleep (delta sleep, stages 3 and 4, slow-wave sleep, SWS) (Beersma and Hoofdakker 1992; Merica and Gaillard 1991), however. Restriction of REM appears to be inextricably linked to alterations of SWS, so that REM SD does not produce a pure alteration in REM sleep to be related to the therapeutic effects of SD.

Studies of catecholamine metabolites and hypothalamo-pituitary-adrenal axis function after SD in depressed patients showed decreased catecholaminergic turnover in SD responders (Gerner et al 1979; Post et al 1976). Cerebrospinal fluid (CSF) levels of methoxyhydroxyphenethylglycol (MHPG) were decreased after SD in depressed patients whose mood improved (responders) but were increased among nonresponders (Post et al 1976; Gerner et al 1979). Lower baseline CSF homovanillic acid (HVA) levels were found among responders than nonresponders (Gerner et al 1979; Post et al 1976). Plasma corticotropin releasing hormone and adrenocorticotrophic hormone (ACTH) levels declined after SD in depressed subjects, whereas growth hormone levels increased (Reist et al 1991). Among depressed patients who showed a mood improvement after SD, baseline nonsuppression of cortisol release after dexamethasone normalized to a suppressor status (Nasrallah et al 1980; Holsboer-Trachsler et al 1988).

Some antidepressant medications prolonged the improvement of mood in depressed patients after SD suggesting a synergism, possibly via a common neurobiologic mechanism. Clomipramine (Elsenga and van den Hoofdakker 1982/1983; Wirz-Justice et al 1976) and nortriptyline (Shelton 1993), both tricyclic mixed 5-HT and norepinephrine reuptake inhibitors, and lithium (Baxter et al 1986) may prolong the SD-induced remission of depressive symptoms. SD hastened the response to clomipramine in patients with depression (Loosen et al 1976). In several patients from one case series (Wirz-Justice et al 1976) but not others (Hochli et al 1986; Riemann and Berger 1989), clomipramine was more effective as an antidepressant in patients whose mood improved after SD than in patients who did not improve. Further studies of SD interactions with antidepressant medications are needed (Liebenluft and Wehr 1992).

Preclinical (Blier et al 1986; Fuxe et al 1984; Blier et al 1990) and clinical (Charney et al 1984; Price et al 1989a, 1989b, 1990; Elphick et al 1990; Blier et al 1987; Heninger

and Charney 1987; Cowen et al 1989) evidence suggests that the therapeutic action of most antidepressant medications in the treatment of depression may involve an enhancement of 5-HT function. The intravenous administration of tryptophan (TRP), the precursor of 5-HT, causes plasma PRL levels to rise, and this response is used as an index of 5-HT function. Reviews of previous work suggest that this rise is blunted in patients with depression, and that it normalizes after chronic administration of most antidepressant treatments (Upadhyaya 1991; Delgado and Charney 1991).

The present investigation examined the effects of SD on the prolactin (PRL) rise produced by intravenous administration of TRP in depressed patients. We predicted that, similar to many other antidepressant treatments, SD would enhance the 5-HT-mediated hormonal response.

Methods

Patient Selection

Eleven medication-free depressed patients between the ages of 18–55 (44.3 ± 14.6 ; this and all following variances expressed as mean and SD); the age of six men (47.0 ± 7.5) was greater than five women (34.4 ± 18.8 , NS). Three men and one woman were military veterans; there were three inpatients and eight outpatients. Three patients were medication-naïve, the others were medication free for a minimum of 1 month (mean = 28.2 months, reflecting recurrent depressions with long periods between episodes). Patients were admitted to the study after a complete medical and psychiatric history, a physical and neurological exam, and thorough explanation of the procedures with signed informed consent. Depressed patients all met Diagnostic and Statistical Manual III-Revised (DSM-III-R) criteria for Major Depressive Episode and were rated on a semi-structured 25-item version of the Hamilton Depression Rating Scale (HDRS). Depressive symptoms were assessed by experienced raters with established reliability on the HDRS.

Positive urine screen or ongoing drug or alcohol use was cause for exclusion. Patients with a history of previous drug or alcohol use stated that they were substance-free for at least 3 months and alcohol free for 3 weeks at the time of the first study. No patient had a serious medical or neurological illness, or abnormality on routine laboratory studies. Female patients with childbearing potential were tested for pregnancy less than 7 days before beginning the first infusion test, and were not studied unless using a barrier method of birth control. Patients with a history of mania or psychosis, and patients with current panic or seizure disorders were excluded.

Overview of Procedures

A 2-week, crossover study design was used. Drug-free depressed patients were tested after both SD and undisturbed sleep (US) in random order. Two 3-day tests (pretest baseline day with the SD or US night, TRP infusion day, and follow-up day) were separated by 1 week.

Undisturbed Sleep

Undisturbed, spontaneous sleep was monitored for approximate duration between 10:15 PM and 7 AM by observing patients every 15 min. Patients were instructed to signal with a hand gesture only if they felt fully awake when observers passed. Sleep time was counted as 15 min for each observation at which the patient did not wave. Patients who slept for less than 4 hr would not have been tested the following day but this did not occur in this sample.

Sleep Deprivation

For SD, patients were not allowed to sleep or even nap briefly for the entire night. This was monitored under close observation of nursing staff. Patients used ward recreation facilities in well-lighted rooms and fasted after 11 PM.

Rater Blind

Patients and ward nursing staff were aware of the assignment to US or SD. The clinicians, nurses, research assistants, and investigators who administered clinical scales, drew the blood samples, and infused TRP were blind to the sleep condition, however. Plasma samples were coded to ensure that laboratory technicians were also blind to condition.

TRP Infusion

TRP for intravenous infusion (Sigma, Inc., St. Louis, MO) was prepared as a 14 mg/ml pyrogen-free solution in 500 cc of saline at pH 7.4, filter-sterilized and tested for pyrogenicity and sterility before use.

After fasting overnight, patients went to the Biological Studies Unit at 8 AM and adapted to the surroundings in a supine position. During the entire 4 hr test, patients remained awake, in a supine position with their heads elevated. At 8:30 AM a catheter was placed in the antecubital fossa. Sixty minutes later, TRP (100 mg/kg) was administered over a 20-min period. Blood samples were drawn from a small intravenous catheter that was in place for the duration of testing on day 2 and kept patent by a slow infusion of normal saline. Samples were drawn for PRL levels on the baseline day (day 1) at 9 AM, on day 2 just prior to the infusion and then every 30 min for 3 hr, and again at 9 AM on the third day.

Mood Ratings

Changes in mood after SD ratings were assessed using a modified HDRS minus the sleep items at four time points: the pretest baseline day (9 AM on day 1), preinfusion (9 AM on day 2), postinfusion (3 PM on day 2), and at follow-up (9 AM on day 3).

Prolactin Measurements

Plasma was separated by centrifugation immediately after each sampling time, frozen on dry ice, and stored at -20°C for radioimmunoassay (GammaDab, Incstar Corp, Stillwater MN). Interassay and intraassay coefficients of variation were 3.4 to 10.5% and 4 to 6.6%, respectively. Duplicate samples from multiple tests in the same patient were assayed together and showed high correlation between duplicates ($r^2 = 0.987$, $p = 0.0001$).

Statistical Analyses

HDRS scores were analyzed by repeated measures analysis of variance (ANOVA), and ANCOVA (SPSS for Mac-Intosh, SPSS Inc, Chicago) using pretest baseline scores on day 1 as the covaried term. Decreases in HDRS scores of 25% from baseline to day 2 defined a positive mood response.

Changes in prolactin levels from preinfusion (day 2) baselines to peak levels postinfusion were compared between the US and SD tests using the paired Student's *t*-test. For analyses of hormone levels the ANOVA within subjects with repeated measures was used to compare SD and US. Correlations among baseline clinical state (HDRS score), clinical response (HDRS change) and prolactin levels were tested using the Pearson correlation coefficient. Other comparisons (Pearson correlations) were run for effects of age, pretest weight loss on baseline PRL, and peak change in PRL level. All outcome analyses were performed as two-tailed tests, with significance accepted at the $p < 0.05$ threshold.

Results

HDRS Scores

There was no significant change in mean HDRS scores from baseline (day 1) to after SD (ANOVA: $df = 10,33$; $F = 2.35$; $p = 0.09$; NS) nor were changes from baseline to scores following SD different from changes following US ($df = 10,1$; $F = 0.837$; $p = 0.38$; NS) (Table 1). Positive mood responses were observed in four patients after SD and four patients after US; three of whom had mood responses after both of the tests. Patients who responded to SD described a reduced depressed mood, without hypomanic or unusual

Table 1. Hamilton Depression Rating Scale Scores^a

Parameters	Sleep deprived	Undisturbed sleep
All patients (n = 11)		
Day 1	28.5 ± 10.6	26.9 ± 11.3
Day 2, AM	25.4 ± 13.2	23.8 ± 10.7
Day 2, PM	22.5 ± 12.2	24.3 ± 10.9
Day 3	24.9 ± 11.8	23.5 ± 10.1
Female patients (n = 5)		
Day 1	33.0 ± 10.2	28.4 ± 15.7
Day 2, AM	28.4 ± 17.7	27.8 ± 11.5
Day 2, PM	28.4 ± 13.8	27.4 ± 12.0
Day 3	29.5 ± 13.0	27.8 ± 12.9
Male patients (n = 6)		
Day 1	24.8 ± 10.3	25.7 ± 7.3
Day 2, AM	22.8 ± 9.1	20.5 ± 9.7
Day 2, PM	17.7 ± 9.0	21.7 ± 10.1
Day 3	21.2 ± 10.6	19.3 ± 4.6

^aDepression ratings for two tests, a week apart, prior to sleep condition (day 1), after sleep condition but prior to tryptophan infusion (day 2, AM), 5 h after tryptophan infusion (day 2, PM), and after recovery sleep (day 3). No significant changes were observed between tests or during tests. (mean ± standard deviation)

symptoms. Both responders and nonresponders to SD complained of sleepiness.

Prolactin Levels

There was a robust effect of SD on the peak change in prolactin levels from preinfusion (day 2 baseline) level to the highest postinfusion level in female patients (Table 2, Figure 1). In all five female patients, TRP-induced increases in PRL were greater following SD than US. However, (n = 5), but not men (n = 6), showed significantly greater mean changes in PRL release in response to TRP infusion following SD in comparison to US (paired *t*-test, *p* < 0.005). Mean scores for all 11 patients together remained significant (*p* < 0.05). Preinfusion baseline PRL levels were unchanged between post-SD and post-US (day 2) test-

Table 2. Serum Prolactin Levels^a

Parameters	Day 2 baseline	Day 2 peak	Peak change from baseline
All patients (n = 11)			
US	5.41 ± 1.35	7.87 ± 3.04	2.49 ± 2.72
SD	5.51 ± 1.38	12.32 ± 5.85	6.81 ± 5.15 ^b
Females only (n = 5)			
US	6.36 ± 1.50	8.02 ± 3.67	1.66 ± 2.78
SD	6.30 ± 1.49	17.52 ± 3.45	11.22 ± 3.40 ^c
Males only (n = 6)			
US	4.62 ± 0.40	7.98 ± 3.03	3.18 ± 2.72
SD	4.85 ± 0.96	7.75 ± 2.76	3.13 ± 2.86

^aProlactin responses after tryptophan infusion after undisturbed sleep (US) or sleep deprivation (SD). (mean ± SD in ng/mL).

^bDifferent from US by paired, two-tailed Student's *t*-test (df = 10, *t* = 2.42, *p* < 0.05).

^cDifferent from US by paired, two-tailed Student's *t*-test (df = 4, *t* = 9.56, *p* < 0.005).

ing. There was no significant correlation (Pearson) between peak changes in PRL and HDRS score changes or between peak changes in PRL and age.

Discussion

The primary findings of the current investigation are that SD enhances the PRL response to TRP and that this is observed in female but not male depressed patients. The data is consistent with previous work demonstrating that in depressives, many, but not all, antidepressant treatments enhance the rise in serum PRL levels induced by TRP.

The gender data may be viewed in the context of several previous investigations. Healthy female subjects have consistently shown significantly greater PRL responses (peak minus baseline) to TRP infusion (Heninger et al 1984; Delgado et al 1989; Charney et al 1988) and other 5-HT agonists (e.g., *d,l*-fenfluramine, McBride et al 1990) than healthy males. In studies with smaller sample sizes, healthy women tended to show greater PRL responses to TRP infusion than men, but without reaching significance (MacIndoe and Turkington 1973; Charney et al 1982; Goodwin et al 1987; Cowen and Charig 1987; Upadhyaya et al 1991). Reports of TRP infusion effects on PRL release in drug-free depressed patients suggest that women have greater PRL changes than men (Heninger et al 1984), although significance levels for this difference were not always reported (Price et al 1991; Upadhyaya et al 1991) or the differences were pronounced only among melancholic patients (Maes et al 1989). Greater PRL responses to TRP in women have also been reported in obsessive-compulsive disorder (Charney et al 1988). In two studies, both women and men depressed patients responded to TRP infusion with less PRL rise than healthy subjects (Heninger et al 1984; Cowen and Charig 1987), and within depressive men and women, this blunted response was state-dependent with normal responses following prolonged remission (Upadhyaya et al 1991). Two other studies found differences between female healthy subjects and depressed patients but not between the male patients and their counterparts (Maes et al 1989; Price et al 1991), however. In summary, women consistently show PRL responses that are larger than men, and blunted PRL release is consistently observed in severely depressed women; also, findings in male patients vary among studies. This suggests that the PRL response following TRP infusion may not have an adequate range to show significant change. Consistent with these assertions, the present data suggest that the effects of 5-HT agonists on PRL secretion may be an insensitive index of changes in 5-HT function in depressed men.

An association between mood improvement and increased PRL response to TRP was not shown in this small sample. All four SD mood-responders (in whom HDRS scores decreased by >25% on HDRS) also showed strong

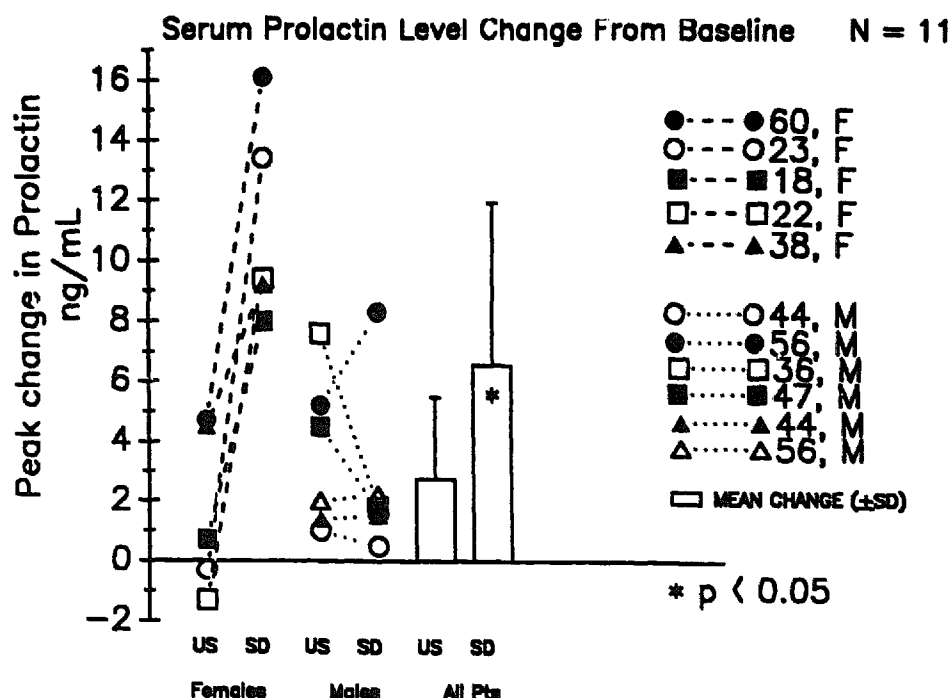


Figure 1. Changes of individual patients in the prolactin responses (peak following infusion minus preinfusion) to infused tryptophan after undisturbed sleep (US) or sleep deprivation (SD). The dashes (women) or dots (men) connect changes shown by individuals from one test to the other. Tests were in random order, 1 week apart (see Methods).

PRL responses (≥ 8 ng/mL rise from baseline) whereas only three of eight mood nonresponders showed this prolactin response, however. A larger sample is needed to more definitively determine whether there is a relationship between SD related mood improvement and enhanced 5-HT function. It may be that SD increases 5-HT function generally, but only patients with a preexisting abnormality in 5-HT function experience mood elevation.

PRL release, both at baseline and in response to 5-HT agonists, is sensitive to numerous factors that may be altered after SD. The spontaneous nocturnal and early morning rise, and slow late morning decline in PRL in healthy subjects (Parker et al 1973) might be phase-shifted (thereby showing lower levels) after SD in depressed patients (Desir et al 1982). In healthy subjects, spontaneous (nonstimulated) prolactin release rises and peaks during the early morning hours and this rise was enhanced after SD in one study (Baumgartner et al 1990) but not another (Opstad 1991). In either case, this is not a likely cause of the altered release in response to the 5-HT stimulus as mean baseline PRL levels in our data were virtually identical after SD and US.

PRL release varies during menstrual cycles, whether spontaneously (Jernstrom et al 1992), or in studies using d-fenfluramine (O'Keane et al 1991) or buspirone (Dinan et

al 1990) challenges. Premenstrual (luteal phase) tests show higher PRL release than tests during the follicular phase in these studies. A difference in timing of the presently reported tests through menstrual cycles does not appear to explain the strong effect of SD on PRL release in the present sample since, of five women, three were not menstruating, and one was given both tests in the preovulatory phase.

This study presents evidence that SD, which is associated with an improvement of mood in a subpopulation of depressed patients, may enhance 5-HT function as measured by the PRL response to intravenous TRP. This observation was confined to female depressed patients, however. This suggests that future studies of tryptophan infusion effects on prolactin release must carefully evaluate the effects of gender, and that new methods for assessing 5-HT function in males are needed. Further study of the mechanisms by which SD increases 5-HT function may lead to more rapid treatments for mood disorders.

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